

Section A — Concept Application

Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

ANS: Here is how to select and confirm compatibility in brief:

- **CPU & Motherboard:** Match the physical socket (e.g., Intel LGA 1700 or AMD AM5) and ensure the chipset supports the processor generation.
- **RAM:** Ensure the memory type (DDR4 or DDR5) and slot form factor (UDIMM) match both the motherboard and CPU specs.
- **Storage:** Verify the motherboard has an M.2 slot supporting the drive's protocol (NVMe PCIe vs. SATA).

To Confirm: Check the motherboard manufacturer's official CPU Support List and Memory QVL (Qualified Vendor List), or run your exact parts through PCPartPicker.

Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

ANS: Storage Comparison

- **HDD:** Mechanical spinning platters (~150 MB/s). Extremely slow, cheap storage.
- **SATA SSD:** Flash memory using SATA cable (~550 MB/s). Moderate speed, silent.
- **NVMe SSD:** Flash memory attached directly to PCIe lanes (3,500–7,500+ MB/s). Fastest speed, lowest latency.

Recommendation: NVMe M.2 SSD

Why: Higher IOPS (Input/Output Operations Per Second) and low latency eliminate bottlenecks during multi-tasking, providing near-instant OS boot, rapid file transfers, and minimal application load times.

Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.

ANS: GPU Role

- **Offloads Visual Tasks:** Handles display outputs, multi-monitor setups, and hardware acceleration for web browsers and video calls, freeing up the CPU.
- **Integrated is Enough:** Standard office workloads only require an integrated GPU (iGPU); discrete GPUs are reserved for heavy 3D/video work.

Cooling System Role

- Prevents Thermal Throttling: Keeps temperatures low so the CPU doesn't drop clock speeds to protect itself.
- Ensures Stability & Lifespan: Stops heat-induced crashes and protects silicon from long-term hardware degradation under continuous loads.

Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

ANS: Hardware compatibility ensures all parts fit physically, communicate electronically, and receive correct power to operate correctly.

Key Risks of Incompatible Parts

- Physical Mismatches: Components fail to fit (e.g., DDR5 RAM won't fit a DDR4 slot; oversized GPU won't fit the case).
- Failure to Boot: Mismatched sockets (e.g., Intel CPU on an AMD motherboard) or outdated BIOS prevent the PC from turning on.
- Crashes & Instability: Unsupported RAM speeds or incorrect voltages cause frequent Blue Screens (BSOD) and file corruption.
- Hardware Damage: Incorrect power supply cabling or improper pin connections can short-circuit components.
- Performance Bottlenecks: Modern parts drop to legacy speeds if connected to outdated slots (e.g., PCIe Gen 4 SSD in a Gen 3 slot).

Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on

ANS: Choosing the PSU

- Wattage: Add a 20–30% safety buffer over total component TDP (400W–500W is ideal for standard office builds; 650W+ if adding a discrete GPU).
- Efficiency: Pick an 80 PLUS Bronze or Gold certified unit from a reputable brand.

Pre-Power-On Checks

- Power Connections: Ensure the 24-pin ATX, 8-pin CPU, and SATA/PCIe cables are fully seated until latches click.
- Short-Circuit Prevention: Verify brass motherboard standoffs are correctly placed and no loose screws touch the circuit board.

- **Physical Safety:** Ensure all fan blades are clear of stray cables and the PSU toggle switch is set to OFF (0) before plugging into a grounded outlet.

Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

ANS: Confirming Successful POST

- **Visual Output:** The display turns on and shows the BIOS splash screen.
- **Diagnostic LEDs:** Motherboard debug LEDs (CPU, DRAM, VGA, BOOT) cycle through and all turn **OFF**.
- **Peripherals:** Keyboard lights illuminate and fan speeds drop from full speed to quiet idle.

If POST Fails

1. **Read Debug LEDs:** Identify which status LED stays lit (CPU, DRAM, or VGA).
2. **Reseat RAM:** Re-insert memory sticks firmly into recommended slots until latches click.
3. **Check Power Cables:** Verify 24-pin ATX and 8-pin CPU connectors are fully seated.
4. **Clear CMOS:** Remove the motherboard coin battery for 5 minutes to reset BIOS settings.
5. **Minimal Boot:** Disconnect non-essential hardware, running only 1 RAM stick, CPU, and PSU.

Section B — Practical Coding Tasks

Task 1: Component Identification & Functions

ANS:

Component	Primary Function	Key Specification
Motherboard	Connects all hardware and routes power and data.	Socket & Chipset (e.g., LGA 1700, AM5)

CPU	Executes instructions and processes system logic.	Cores, Threads & Clock Speed (GHz)
RAM	Provides fast temporary memory for active apps.	Capacity & Type (e.g., 16GB DDR5)
GPU	Renders visual outputs and graphics tasks.	Model or iGPU type & VRAM
Storage	Retains OS, apps, and files permanently.	Interface & Capacity (e.g., 1TB NVMe M.2)
Cooling	Dissipates CPU heat to prevent throttling.	Type (Air or AIO Liquid) & Fan size

Task 2: Motherboard Ports & Socket Identification

ANS:

Internal Sockets & Slots

- **CPU Socket:** Houses the processor (e.g., LGA 1700, AM5).
- **DIMM Slots:** Holds RAM modules (DDR4/DDR5).
- **PCIe x16 Slot:** Connects primary expansion cards (GPU).
- **PCIe x1/x4 Slots:** Connects smaller add-on cards (Wi-Fi, sound).
- **M.2 Slots:** Secures fast NVMe SSD storage.

Internal Power & Headers

- **24-Pin ATX:** Main motherboard power input.
- **8-Pin EPS:** Dedicated CPU power input.
- **SATA Ports:** Connects standard 2.5"/3.5" drives.
- **Front Panel Header:** Connects case power button, reset switch, and LEDs.

Rear I/O Ports

- **HDMI / DisplayPort:** Video outputs for monitors.
- **USB Ports (Type-A / Type-C):** Connects peripherals and external storage.
- **Ethernet (RJ-45):** Wired network connection.
- **3.5mm Audio Jacks:** Microphone, headphone, and speaker connections.

Task 3: RAM & Storage Installation

ANS:

ESD Safety: Unplug power cable, hold PC power button 5s to drain residual power, and touch bare case metal to ground yourself.

RAM:

- *Remove:* Push down slot side-latches to eject module.
- *Reinstall:* Align bottom notch with slot key; press firmly down until latches click shut.

Storage:

- *NVMe M.2:* Remove mounting screw, pull drive out at 30° angle. Reinsert at 30°, press flat, tighten screw.
- *SATA:* Unplug power/data cables. Reinstall by plugging both cables back in flush.

Confirmation: Boot into BIOS (**Del/F2**) and verify RAM capacity and Storage drive model display under System Summary.

Task 4: Full Desktop PC Assembly

ANS:

1.Preparation & Safety:Workspace & Anti-Static.

- Set up on a clean, non-conductive surface.
- Ground yourself by touching bare metal or wearing an anti-static wrist strap.

2.Out-of-Case Installation:Core components installed on motherboard first.

- **CPU:** Unlatch socket, align golden corner triangle, lower gently, and lock lever.
- **RAM:** Align stick notch with slot key; press firmly down until both side latches click shut.
- **M.2 SSD:** Insert into slot at 30° angle, press flat, and secure with mounting screw.
- **CPU Cooler:** Apply thermal paste, secure cooler in a cross pattern, and connect fan to **CPU_FAN**.

3.Case Mounting:Securing hardware into chassis.

- **PSU:** Install power supply into bottom/rear chamber and secure with 4 screws.
- **Motherboard:** Verify standoffs align, place board, and tighten mounting screws.

4. Cabling & GPU Installation: Power & Expansion.

- Connect **24-pin ATX**, **8-pin CPU**, and **Front-Panel** cables securely.
- **GPU:** Remove case slot covers, press GPU into top PCIe x16 slot until latched, screw into frame, and connect PCIe power cables.

5. Power On & POST Verification: Initial Boot.

- Plug display cable into GPU (not motherboard), connect power, and press case power button.
- Verify fans spin, motherboard debug LEDs cycle off, and BIOS splash screen displays.

Section C — Mini Capstone Project

Mini Project: Workstation Deployment & Boot-Failure Diagnosis

ANS:

1. Top 3 Boot-Failure Diagnostics

- **RAM:** Unseated modules block POST, lighting DRAM debug LEDs or causing continuous reboot loops.
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- **Power Cables:** Loose 24-pin ATX or 8-pin CPU cables prevent motherboard ignition.
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- **CPU & Socket:** Bent pins, improper seating, or missing **CPU_FAN** connections trigger instant safety shutdowns.
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2. Motherboard–CPU Compatibility

- **Socket Match:** Physical pins must match (e.g., Intel LGA 1700 vs. AMD AM5).
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- **BIOS Firmware:** Matching sockets still require updated motherboard BIOS microcode to recognize newer CPU generations.
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3. RAM & Storage Verification

- **Physical Check:** Confirm RAM clips clicked shut and M.2 SSDs are screwed down flat.
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- **BIOS Confirmation:** Verify total memory capacity (e.g., 16GB) and drive model names show in the BIOS summary.
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4. Deployment Checklist

1. **Out-of-Case POST Test:** Verify basic boot outside the chassis before final installation.
- 2.
3. **Standoff Alignment:** Ensure brass standoffs strictly align with motherboard holes to prevent shorts.

- 4.
5. **Latch Verification:** Check all power cables (24-pin, 8-pin EPS) for audibly clicked latches.
- 6.

5. SSD vs. HDD Justification

- **Speed:** NVMe SSDs (3,000+ MB/s) eliminate system lag and boot apps instantly compared to slow HDDs (150 MB/s).
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- **Reliability:** Zero moving parts eliminates mechanical failure risks and noise during heavy workstation use.

Section D — AI-Augmented Learning

Step 1: AI-Generated Diagnostic Checklist



STEP 1: PSU & Electrical Check

- AC cord plugged in & PSU rocker switch set to ON (I).
- Internal 24-pin ATX and 8-pin CPU cables fully latched.

STEP 2: Motherboard Diagnostics

- Check Debug LEDs (CPU, DRAM, VGA, BOOT).
- Verify case Front-Panel POWER SW cable connection.

STEP 3: Memory Diagnostic (RAM)

- Reseat RAM in primary slots (A2/B2) until latches click.
- Test boot with a single RAM stick in slot A2.

STEP 4: Storage Diagnostic

- Confirm M.2 SSD is seated flat and screwed down.
- Enter BIOS (Del/F2) to verify drive detection.

ABC Solutions Application: Step 3 catches unseated RAM or improper slot placement—the primary cause of fans spinning with no display on newly assembled lab PCs.

Step 2: Manual Corrections (Safety & Logic Fixes)

- **Missing ESD/Power Drain Precaution (Safety Flaw):** Step 3 instructed opening the case and reseating RAM without safety protocols.
- - *Fix:* Insert mandatory step: **Unplug AC power, hold case power button 5s to drain capacitors, and ground self on metal chassis before touching internal components.**
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- **Out-of-Order Front Panel Check (Logic Flaw):** Step 2 placed **POWER SW** checks after debug LEDs, but a loose power switch prevents the PC from attempting to turn on at all.
- - *Fix:* Move **Front-Panel POWER SW verification** directly to Step 1 alongside PSU checks.